* Digital Communications *

SALEH RAMADAN

"Computer"

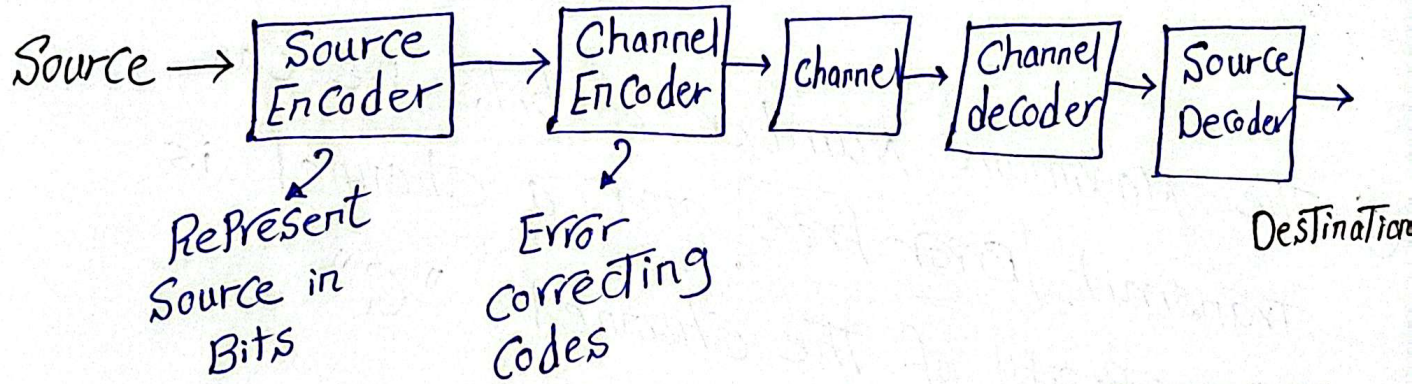
"Section 1"

"Summary"

*** Communication:** Tx → Channel → Rx

We have "Info" at Tx, and need Rx knows it

*** Block Diagram:**



*** Goal:** Maximize Rate of data to serve more Sources

1. The Number of bits ~~Per s~~ **Produced** Per second Should be **Minimized**
2. The Number of bits **Transmitted** Per second Should be **maximized**

CamScanner

②

*

Information Theory

- Source Encoding → Entropy "H" → Function of Source
- Channel Encoding → Capacity "C" → Function of Channel

① The Minimum Number of bits required to represent a source is The Entropy of Source "H"

② The Maximum Number of bits we can transmit **error free** over a channel is The capacity of The Channel "C"

Then $H \leq C$

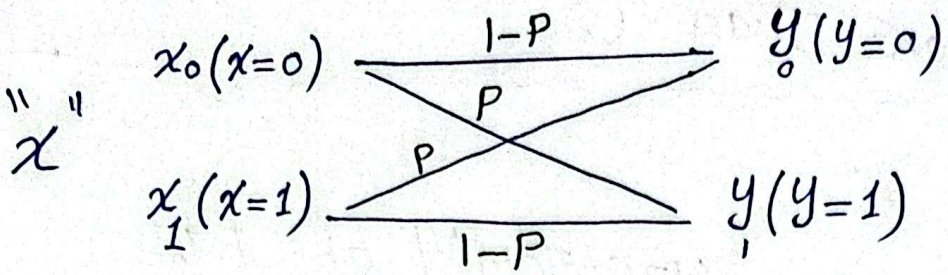
---

* NOTES:

① A Source is a random, Produces random symbols
→ Random variable

② A Channel is a Probablistic mapping between input and output.

* Binary symmetric Channel:



$$P(y|x) = \begin{matrix} & y_0 & y_1 \\ x_0 & P(y_0|x_0) & P(y_1|x_0) \\ x_1 & P(y_0|x_1) & P(y_1|x_1) \end{matrix}$$

*Note: $P(x|y) = \frac{P(y|x)P(x)}{P(y)}$

* $P(y_0|x_0) = 1-P$, $P(y_0|x_1) = P$

* $P(y_1|x_1) = 1-P$, $P(y_1|x_0) = P$

* $P(y_0) = P(y_0|x_0)P(x_0) + P(y_0|x_1)P(x_1)$

* $P(y_1) = P(y_1|x_0)P(x_0) + P(y_1|x_1)P(x_1)$

---

* Source Encoding: represent source in efficient manner

"Compression" → Lossless (ZiP, winrar)

"Compression" → Lossy (Voice compression, Video quality)

For Lossless compression

We can't represent source in bits less than "H" ⟹ So we need to study ⟹ Entropy "H"

we also can't transmit bits over the Channel more than C

*Entropy "H":

* It represents average uncertainty of RV.
* It represent amount of Info. in Source
* It represents amount of Surprise.
* It represent min average Number of bits to represent a Source

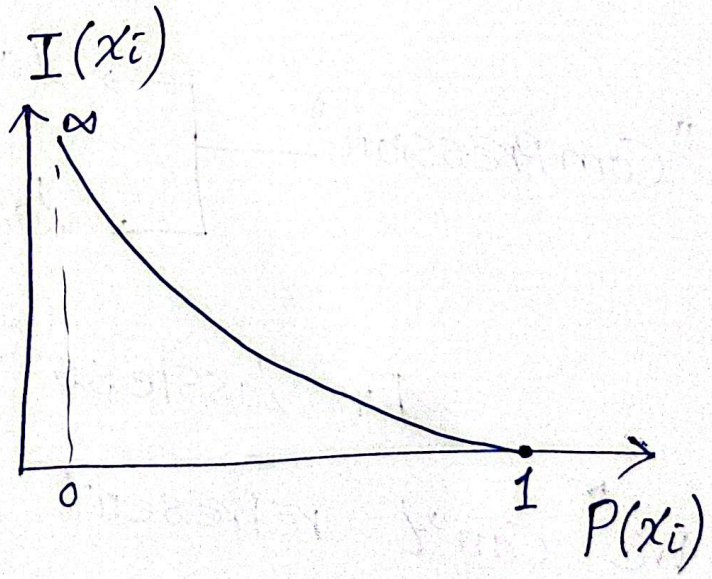
→ If we have RV: $X = \{x_0, x_1, \dots, x_N\}$

$I(x)$ = Self Information

N = # of symbols, $N = 2^I$

$$P(x_i) = \frac{1}{N} = \frac{1}{2^I}$$ "Equally Probable"

$$I = -\log_2 (P(x_i))$$

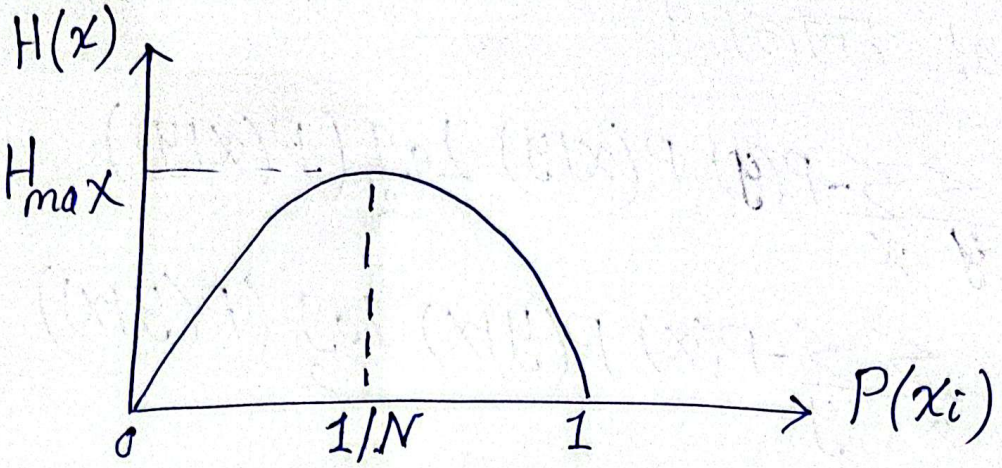


$P = 1 \Rightarrow$ Zero Info
(Certain) (No Surprise)

$P = 0 \Rightarrow \infty$ Info
(Uncertainty) (Huge Surprise)

⑤

$$H(x) = \sum_i - P(x_i) \log_2 (P(x_i)) \quad \text{"bits"}$$



$$0 \leq H \leq H_{max}$$

$$H_{max} = \log_2(N) \quad @ \quad P(x_i) = \frac{1}{N}$$

* Remember: $P(x,y)$: Joint Probability

$$P(x|y) = \frac{P(x,y)}{P(y)}$$

If x, Y are Independant: $P(x,y) = P(x) * P(y)$

$$P(x|y) = P(x)$$

* Joint Entropy: "$H(x,y)$"

$$H(x,y) = \sum_x \sum_y - P(x,y) \log_2 (P(x,y))$$

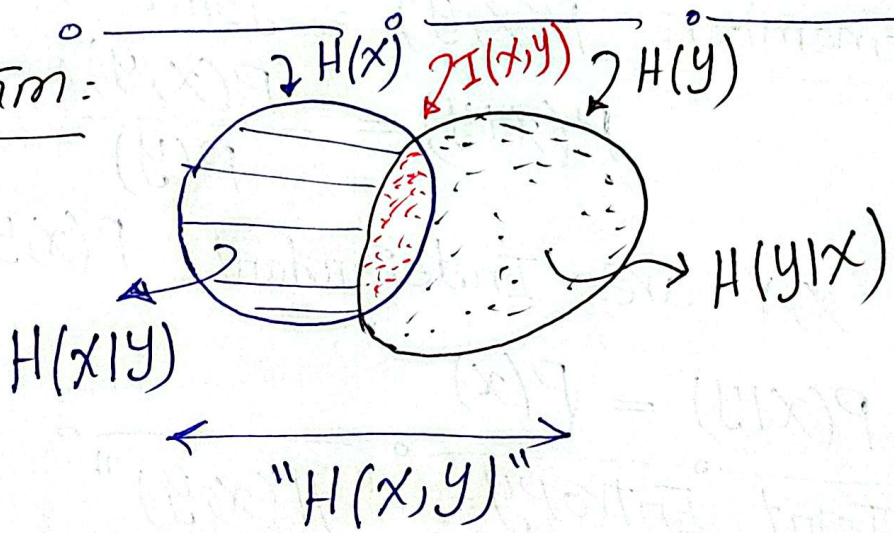
$$H(x,y) \leq H(x) + H(y)$$

If x, y are Independant $\Longrightarrow H(x,y) = H(x) + H(y)$

* Conditional Entropy:

- $H(x|y) = \sum_{y} \sum_{x} -P(y)\, P(x|y) \log_2 (P(x|y))$
- $H(y|x) = \sum_{x} \sum_{y} -P(x)\, P(y|x) \log_2 (P(y|x))$
- $H(x|y) \neq H(y|x)$
- $H(x,y) = H(x) + H(y|x)$
- $H(x,y) = H(y) + H(x|y)$

* Venn Diagram:



* Mutal Information: $I(x,y)$

* It's Info That I can Know about x from y

$$I(x,y) = H(x) - H(x|y) = H(y) - H(y|x)$$

* Capacity "C": Shannon

$C = \max(I(x,y))$ ← Remember

we choose $P(x)$ to maximize $I(x,y)$

$\frac{dI(x,y)}{dP(x)} = 0$

$\log_2(x) = \frac{\ln(x)}{\ln(2)}$